

MULTISPECTRAL EMISSION OF THE SUN DURING THE FIRST WHOLE SUN MONTH: MAGNETOHYDRODYNAMIC SIMULATIONS

ROBERTO LIONELLO¹, JON A. LINKER¹, AND ZORAN MIKIĆ¹

Science Applications International Corporation, San Diego, CA 92121-1578, USA; lionellor@saic.com, linkerj@saic.com, mikicz@saic.com

Received 2008 May 15; accepted 2008 September 3; published 2008 December 1

ABSTRACT

We demonstrate that a three-dimensional magnetohydrodynamic (MHD) simulation of the corona can model its global plasma density and temperature structure with sufficient accuracy to reproduce many of the multispectral properties of the corona observed in extreme ultraviolet (EUV) and X-ray emission. The key ingredient to this new type of global MHD model is the inclusion of energy transport processes (coronal heating, anisotropic thermal conduction, and radiative losses) in the energy equation. The calculation of these processes has previously been confined to one-dimensional loop models, idealized two-dimensional computations, and three-dimensional active region models. We refer to this as the thermodynamic MHD model, and we apply it to the time period of Carrington rotation 1913 (1996 August 22 to September 18). The form of the coronal heating term strongly affects the plasma density and temperature of the solutions. We perform our calculation for three different empirical heating models: (1) a heating function exponentially decreasing in radius; (2) the model of Schrijver et al.; and (3) a model reproducing the heating properties of the quiet Sun and active regions. We produce synthetic emission images from the density and temperature calculated with these three heating functions and quantitatively compare them with observations from EUV Imaging Telescope on the *Solar and Heliospheric Observatory* and the soft X-ray telescope on *Yohkoh*. Although none of the heating models provide a perfect match, heating models 2 and 3 provide a reasonable match to the observations.

Key words: MHD – Sun: corona – Sun: magnetic fields

Online-only material: color figures

1. INTRODUCTION

The solar corona expands out into interplanetary space and fills the heliosphere as the solar wind. It is the conduit by which many of the effects of solar activity are transmitted to the Earth to produce geomagnetic activity. The structure and dynamics of the solar corona are thus of essential interest in heliophysics, and the design of realistic models of the solar corona is an important aspect of understanding coronal phenomena.

The Sun’s magnetic field is a key ingredient of any useful model of global coronal structure. The line-of-sight component of the magnetic field in the photosphere has been measured routinely for many years from the ground (e.g., at Stanford’s Wilcox Solar Observatory (WSO), the National Solar Observatory (NSO) at Kitt Peak, and Mount Wilson Observatory) and in space from the Michelson Doppler Imager (MDI) aboard *Solar and Heliospheric Observatory* (SOHO). The earliest attempts to model the coronal and heliospheric magnetic field, based on photospheric field measurements, used potential (current-free) magnetic field models such as the potential field source-surface (PFSS) model (Schatten et al. 1969; Altschuler & Newkirk 1969) and the potential field current-sheet (PFCS) model (Schatten 1971). These models and their variants are still widely used today, and, despite their simplicity and limitations, have been successful in reproducing many aspects of coronal observations (Hoeksema et al. 1983; Wang & Sheeley 1988, 1995; Zhao & Hoeksema 1995; Neugebauer et al. 1998).

Ideally, a model should self-consistently reconstruct both the plasma and magnetic fields in the corona and solar wind.

It was recognized very early that the magnetohydrodynamic (MHD) equations could provide a self-consistent description of the solar wind plasma. Models of the idealized structure of coronal holes and streamers have been performed in the MHD approximation for over 30 years (Endler 1971; Pneuman & Kopp 1971; Steinolfson et al. 1982; Washimi et al. 1987; Linker et al. 1990; Wang et al. 1993; Linker & Mikić 1995; Keppens & Goedbloed 1999; Usmanov & Goldstein 2003; Hayashi 2005). However, with idealized magnetic field configurations, these models were only useful for studying generic properties of the solar corona; they could not address specific observations. The models are time dependent, but typically find a solution by integrating to a steady state.

MHD models that include the photospheric magnetic field as a boundary condition have existed for more than a decade (Usmanov 1993, 1996; Mikić & Linker 1996; Linker et al. 1999; Mikić et al. 1999; Roussev et al. 2003; Riley et al. 2006; Cohen et al. 2007). Typically these models have avoided the complicated physics of the transition region (e.g., radiative losses, anisotropic thermal conduction, and coronal heating) by setting the ratio of specific heats γ to a reduced value (a spatially varying γ is sometimes used). While these models retain an energy equation, we refer to them as “polytropic” to explicitly identify this approximation. While polytropic MHD models can address many aspects of coronal physics, the approach has fundamental limitations. For example, the plasma density and temperature contrasts between open- and closed-field regions, and between active regions and quiet Sun, do not match observations.

Full-disk emission images in extreme ultraviolet (EUV) and X-rays provide strong constraints on global coronal structure, and are available from past (Skylab, Yohkoh), present (SOHO, Solar Terrestrial Relations Observatory (STEREO), Hinode), and future Solar Dynamics Observatory (SDO) missions. Emission

¹ Now with Predictive Science Incorporated, San Diego, CA, USA
lionellor@predsci.com, linkerj@predsci.com, mikicz@predsci.com

measurements have been used to constrain coronal heating models, usually by comparison with one-dimensional loop models (see Klimchuk 2006 for a review). The loop models solve the hydrodynamic (HD) equations with realistic energy transport along a given magnetic field line, assuming that the magnetic field is static. A magnetic geometry for the field line can be assumed, or it can be taken from a magnetic field model (a potential field is typically used). The solution of the HD equations along a single loop obviously cannot address the emission from a given region of the Sun, let alone the entire corona. Recently, several authors have combined individual one-dimensional loop models, calculated on selected magnetic field lines, to synthesize emission images: Lundquist et al. (2004) found the X-ray and EUV emission of active region AR8210; Schrijver et al. (2004) constrained the heating models by simulating the appearance of the entire solar corona; and Warren & Winebarger (2006) used a steady, uniform heating mechanism and computed emission for 26 solar active regions.

Polytropic MHD models cannot quantitatively address data from EUV and X-ray emission because of the limited fidelity of the density and temperature in these calculations. For this purpose, it is necessary to include in the model a more accurate equation for energy transport, which, like the one-dimensional loop models, includes thermal conduction along magnetic field lines, radiative losses, and the specification of a heating function (Lionello et al. 2001). Previous two-dimensional coronal MHD models that implement some or all of these effects, including Suess et al. (1996), Endeve et al. (2003), and Sittler et al. (2003). Suess et al. (1999), Li et al. (2004), and Endeve et al. (2004), have also introduced multifluid effects. Three-dimensional models of active regions that include these effects have also been performed (Gudiksen & Nordlund 2005b, 2005a; Mok et al. 2008).

In this paper, we describe the application of a three-dimensional MHD algorithm with realistic energy transport (Mikić et al. 1999) to a self-consistent model of the global corona for a specific time period. We refer to this approach as the thermodynamic MHD model. We have chosen to focus on the first Whole Sun Month campaign (1996 August 8 to September 10), a period studied in detail in a special issue of *Journal of Geophysical Research*, and for which Linker et al. (1999) calculated the structure of the corona with a polytropic MHD model. As in the polytropic case, the thermodynamic MHD model self-consistently produces the solar wind, streamer boundaries, coronal holes, and the heliospheric current sheet. We demonstrate by direct comparison with observations that it is possible to reproduce many quantitative aspects of the emissivity of the global corona in X-ray and EUV bands. The assumption of a given specification for coronal heating strongly affects the solutions. While we have not exhaustively compared coronal heating models, we demonstrate the effects of coronal heating with three separate specifications: a simple exponential decay function (model 1), the heating model of Schrijver et al. (2004; model 2), and a locally computed heating model that tries to capture the properties of both the quiet Sun and of active regions (model 3). Synthetic emission images have been calculated using a technique described in Mok et al. (2005) and they have been quantitatively compared with observations from the EUV Imaging Telescope (EIT) on the *SOHO* and the Soft X-ray telescope (SXT) on *Yohkoh*.

This paper is organized as follows: in Section 2 we describe our computational MHD model, we discuss the specification of coronal heating functions, and describe how we calculate the

emission images. In Section 3, we present the results of our investigation. We conclude with a discussion of the relevance of our work.

2. COMPUTING THE SUN'S MULTISPECTRAL EMISSION

We now describe the procedure for computing thermodynamic MHD solutions for the Whole Sun Month time period, the properties of the selected heating models, and the method for calculating emission from the solutions.

2.1. MHD Model with Thermodynamics

The MHD approximation is appropriate for long-scale, low-frequency phenomena in magnetized plasmas such as the solar corona. To reproduce the Sun's emission properties during Whole Sun Month, we have solved the following set of viscous and resistive MHD equations:

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}, \quad (1)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}, \quad (2)$$

$$\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c} = \eta \mathbf{J}, \quad (3)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (4)$$

$$\frac{1}{\gamma - 1} \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = -T \nabla \cdot \mathbf{v} + \frac{m}{k\rho} S, \quad (5)$$

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = \frac{1}{c} \mathbf{J} \times \mathbf{B} - \nabla(p + p_w) + \rho \mathbf{g} + \nabla \cdot (\nu \rho \nabla \mathbf{v}), \quad (6)$$

$$S = (-\nabla \cdot \mathbf{q} - n_e n_p Q(T) + H_{\text{ch}}), \quad (7)$$

where $\mathbf{B}$ is the magnetic field, $\mathbf{J}$ is the electric current density, $\mathbf{E}$ is the electric field, ρ , $\mathbf{v}$, p , and T are the plasma mass density, velocity, pressure, and temperature, respectively, $\mathbf{g} = -g_0 R_\odot^2 \hat{\mathbf{r}}/r^2$ is the gravitational acceleration, η is the resistivity, and ν is the kinematic viscosity. Equation (7) contains the radiation loss function $Q(T)$ as in Athay (1986), n_e and n_p are the electron and proton number density (which are equal for a hydrogen plasma), $\gamma = 5/3$ is the polytropic index, H_{ch} is the coronal heating term (see Section 2.2), and $\mathbf{q}$ is the heat flux. A collisional (Spitzer's law) or collisionless (Hollweg 1978) formulation is used according to the radial distance,

$$\mathbf{q} = \begin{cases} -\kappa_0 T^{5/2} \hat{\mathbf{b}} \hat{\mathbf{b}} \cdot \nabla T & \text{if } R_\odot \leq r \lesssim 10 R_\odot \\ \alpha n_e k T \mathbf{v} & \text{if } r \gtrsim 10 R_\odot \end{cases}, \quad (8)$$

where $\kappa_0 = 9 \times 10^{-7} \text{ erg K}^{-7/2} \text{ cm}^{-1} \text{ s}^{-1}$, $\hat{\mathbf{b}}$ is the unit vector along $\mathbf{B}$, and α is a parameter, which was set to 1. The transition between the two forms occurs smoothly between $7.5 R_\odot$ and $12.5 R_\odot$ (Mikić et al. 1999). The wave pressure term p_w in Equation (6) represents the contribution due to Alfvén waves (Jacques 1977). It is evolved using the Wentzel-Kramers-Brillouin (WKB) approximation² for time-space-averaged Alfvén wave energy density ϵ ,

$$\frac{\partial \epsilon}{\partial t} + \nabla \cdot \mathbf{F} = \mathbf{v} \cdot \nabla p_w - D, \quad (9)$$

² In the WKB approximation, it is assumed that waves are propagating over a background fluid that varies slowly in space and time in comparison with the wavelength and period of the waves.

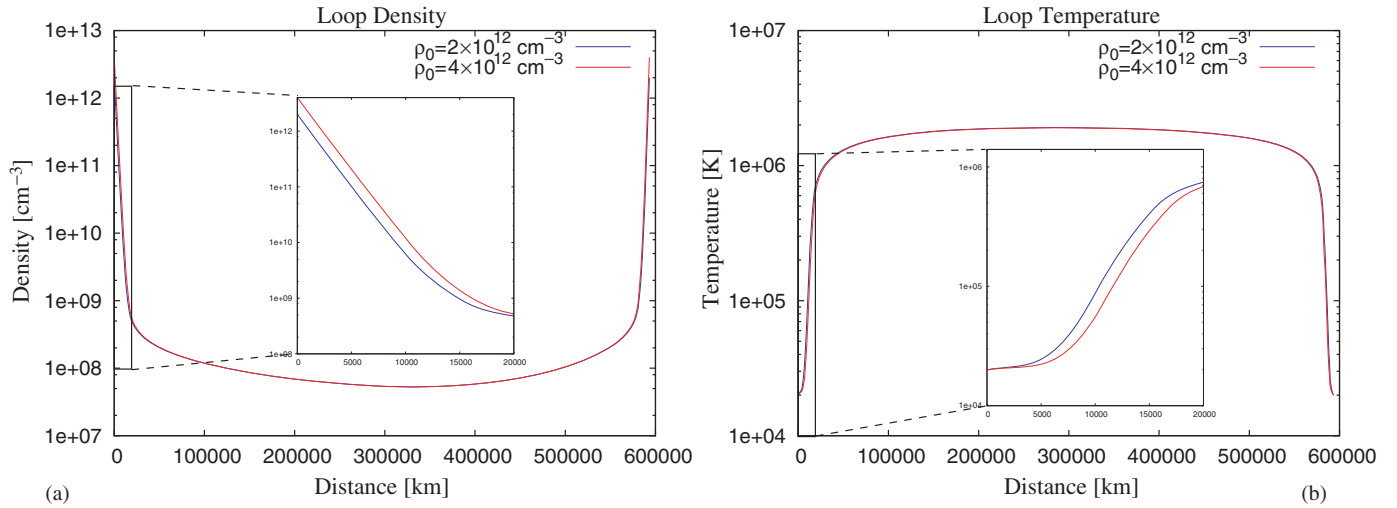


Figure 1. (a) Density and (b) temperature profiles in a one-dimensional loop with the third heating model of Section 2.2. The blue curves show a solution with $\rho_0 = 2 \times 10^{12} \text{ cm}^{-3}$ density as boundary condition at the base, whereas the red curves show the solution when $\rho_0 = 4 \times 10^{12} \text{ cm}^{-3}$. The two solutions are practically indistinguishable in the corona. The location of the loop in the global corona is shown in Figure 2.

(A color version of this figure is available in the online journal.)

where $\mathbf{F} = (3/2\mathbf{v} + \mathbf{v}_A)\epsilon$ is the Alfvén wave energy flux, $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed, and $p_w = \epsilon/2$. The Alfvén wave velocity is $\mathbf{v}_A = \pm v_A \hat{\mathbf{b}}$; in a multidimensional implementation, it is necessary to transport two Alfvén wave fields (waves parallel and antiparallel to $\mathbf{B}$), which are combined to give ϵ . The dissipation term D , which expresses the nonlinear dissipation of Alfvén waves in interplanetary space and is modeled phenomenologically (Hollweg 1978), was set to zero for the present investigation. The boundary conditions on the velocity are determined from the characteristic equations along $\mathbf{B}$. The surface magnetic flux at $r = R_\odot$ is specified from a synoptic magnetic map; for this case we use a smoothed NSO Kitt Peak map for Carrington Rotation (CR) 1913 (1996 August 22 to September 18). At $r = R_\odot$ we also specify the Alfvén wave pressure. At the upper radial boundary, which is placed beyond all critical points, the characteristic equations are used as well.

For the present cases, we have used a nonuniform grid in $r \times \theta \times \phi$ of $131 \times 101 \times 151$ points. The smallest radial-grid interval at $r = R_\odot$ was ~ 300 km; the angular resolution was highest in the area containing the largest active region and the southward extension of the northern coronal hole (which was dubbed “the elephant’s trunk”) and was slightly larger than 1° . A uniform resistivity η was used, corresponding to a resistive diffusion time $\tau_R \sim 4 \times 10^3$ hr, which is much lower than the value in the solar corona. This is necessary to dissipate structures that cannot be resolved since they are smaller than the cell size. The Alfvén travel time at the base of the corona ($\tau_A = R_\odot/V_A$) for $|\mathbf{B}| = 2.205$ G and $n_0 = 10^8 \text{ cm}^{-3}$, which are typical reference values, is 24 minutes (Alfvén speed $V_A = 480 \text{ km s}^{-1}$), so the Lundquist number τ_R/τ_A is 1×10^4 . A uniform viscosity ν is also used, corresponding to a viscous diffusion time τ_ν such that $\tau_\nu/\tau_A = 500$. Again, this value is chosen to dissipate unresolved scales without substantially affecting the global solution. In all the simulations we have used fixed chromospheric values of density and temperature at the base of the domain of $n_0 = 2 \times 10^{12} \text{ cm}^{-3}$ and $T_0 = 20,000$ K, respectively. This is an overestimation of the pressure that does not affect the coronal solution (Mok et al. 2005). In fact, at this value of temperature, radiation loss ($n_e n_p Q(T)$ in Equation 7)

becomes very small. The heat flux also tends to become very small, and a layer with uniform temperature develops near $r = 1R_\odot$, roughly similar to the “temperature plateau” at the top of the chromosphere. The density at the lower boundary is set to a fixed chromospheric value that is large enough (for the specified heating) to produce a temperature plateau. If the specified chromospheric density (and pressure) are too large, the thickness of the temperature plateau will increase slightly (since the scale height in the plateau region is very small, equal to 1200 km at $T = 20,000$ K). If the chromospheric density is too low, the transition region may evaporate. It is not crucial to estimate the required chromospheric density accurately; it is best to overestimate it. An overestimation will simply produce a slightly thicker plateau region, without affecting the overall solution. Figure 1 demonstrates this property by showing one-dimensional loop solutions for different base densities. The coronal parts of the solution are essentially identical. The geometrical properties of this loop were extracted from the three-dimensional solution as described in Section 3. The loop launch point and position in the global corona can be seen in Figure 2.

A principal difficulty in computing three-dimensional coronal solutions using Equations (3)–(7) is related to the extremely steep temperature and density gradients in the transition region, a consequence of the balance between the conduction of heat from the hot corona and radiative loss in the transition region. In order to efficiently model the coupling between the transition region and corona, we have developed a technique that artificially broadens the transition region, while maintaining accuracy in the corona (Z. Mikić et al. 2008, in preparation). We have found that if the temperature dependence of the thermal conductivity $\kappa(T)$ and radiation loss function $Q(T)$ is modified in such a way as to keep the product $\kappa(T)Q(T)$ unchanged, then the coronal solution is not changed significantly. Looking at Equation (7), we see that the balance of radiative losses with thermal conduction in the transition region can be expressed in a dimensional analysis as

$$n_e n_p Q(T) \sim \kappa(T) * T/L^2, \quad (10)$$

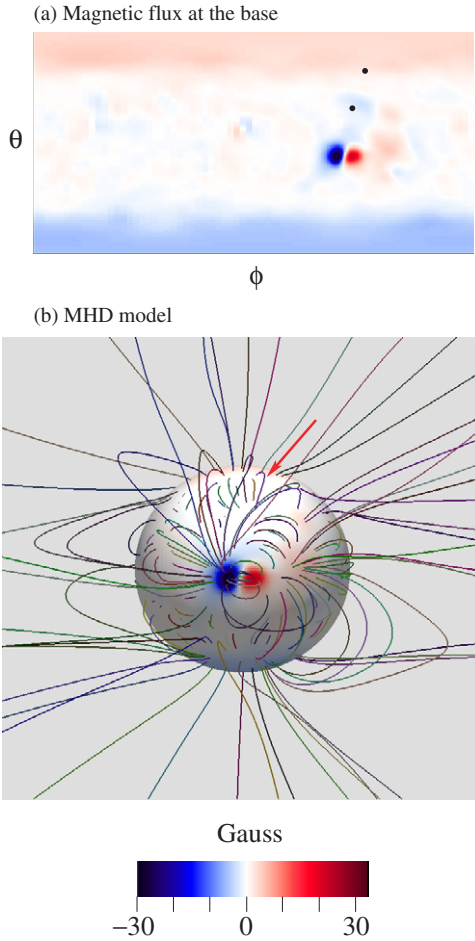


Figure 2. Magnetic field model for the Sun around the time of Whole Sun Month (Carrington Rotation 1913, 1996 August 22 to September 18). (a) The magnetic flux distribution prescribed at the base of the domain and used to calculate the magnetic field configuration for the three heating functions described in Section 2.2. The dots mark the footpoints of the field line used in the 1D loop calculations of Figures 1, 3, and 7. (b) The magnetic flux distribution projected on the solar surface and selected magnetic field lines from the MHD solution. The arrows point to the loop used in the abovementioned one-dimensional solutions.

(A color version of this figure is available in the online journal.)

where L is the length scale associated with the temperature gradient. We see that

$$L \sim \sqrt{\frac{\kappa(T) * T}{n_e n_p Q(T)}}. \quad (11)$$

To broaden the transition region we need to increase κ and decrease Q at low temperatures. We have modified $\kappa(T)$ to be constant for temperatures below $T_c = 500,000$ K. Accordingly, we reduce Q to keep κQ unchanged.

With this modification, the smallest transition region length scale is expected to be 400 km (for a loop $p = 1 \text{ dyn cm}^{-2}$). Indeed, when we simulate a one-dimensional loop with this modification, we find that the transition region is broadened significantly (by a factor of ~ 400), and yet the coronal solution remains virtually unchanged. Figure 3 shows a comparison of loops calculated with and without this modification. The geometry of the loop is the same as that of Figure 1. We find that the emission of the loop in EUV is not significantly modified for coronal temperatures (above $\sim 500,000$ K). More

accurate calculations at lower temperatures can be achieved by lowering the temperature at which the modification occurs, with the consequence of a requirement for higher resolution.

2.2. Heating Functions

In the thermodynamic MHD model we have just described, it is necessary to specify explicitly a coronal heating function $H_{\text{ch}}(r, \theta, \phi)$. The physical mechanism that heats the solar corona and powers its emission in the X-ray and EUV bands continues to be a hotly debated topic of research. Although there is general consensus that what heats the corona must ultimately involve conversion of magnetic energy into heat, it is not clear whether the main mechanism can be identified as the dissipation of high-frequency waves or rather in the slow energy build-up through photospheric motions followed by its rapid release, as proposed by Parker (1979). Mandrini et al. (2000), Priest et al. (2000), and Aschwanden (2001) have summarized the characteristics and physical implications of a number of models that are currently believed to be viable. Besides these physics-based models, phenomenological heating models have also been developed, starting from studies of correlations between radiation losses and magnetic flux in the Sun and other stars (Gurman et al. 1974; Schrijver et al. 1985; Fisher et al. 1998). Pevtsov et al. (2003) concluded that there is a universal linear correlation between magnetic flux and X-ray radiance.

Typically, coronal heating mechanisms have been constrained by calculating the X-ray and EUV emission of coronal loops and comparing it with observations. Plasma loops, first observed in Skylab, can be studied with simple, one-dimensional models. Initially, one-dimensional loops were investigated with static models (Rosner et al. 1978; Craig et al. 1978; Hood & Priest 1979; Vesecky et al. 1979; Serio et al. 1981), and then with numerical HD simulations (Wu et al. 1981; Cheng et al. 1983; Klimchuk et al. 1987; Mok et al. 1991; see also Bray et al. 1991 for a review). From the loop density and temperature distributions provided by the models, the emission (at least for optically thin lines) can be calculated (e.g., Karpen et al. 2001) using the CHIANTI database (Dere et al. 1997). As discussed in the introduction, amalgamations of loop atmosphere calculations for several thousand loops have been used to simulate emission from active regions (Lundquist et al. 2004; Warren & Winebarger 2006). The most comprehensive attempt to date to constrain coronal-heating models with the appearance of the entire corona is that of Schrijver et al. (2004).

For the present study, we show results for three heating models. The heating fluxes, in $\text{erg cm}^{-2} \text{ s}^{-1}$, calculated at the base of the corona for each model are shown in Figure 4.

1. The first heating function we considered is a simple exponential decay law:

$$H_{\text{exp}} = H_0 \exp(-(r - R_{\odot})/\lambda_0), \quad (12)$$

where $H_0 = 4.9128 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ and $\lambda_0 = 0.7 R_{\odot}$. This heating function yields fast wind in one-dimensional models (Withbroe 1988) when used in conjunction with an Alfvén wave pressure at the base of the corona of $p_w = 8.36 \times 10^{-2} \text{ dyn cm}^{-2}$. The total power injected into the corona is $4.93 \times 10^{27} \text{ erg s}^{-1}$.

2. The second model is based on that of Schrijver et al. (2004), who calculated one-dimensional loop solutions based on the following heat flux at the base of the corona:

$$F_{\text{ch}} = 4 \times 10^{14} B/L \exp[-(B/500)^2] \text{ erg cm}^{-2} \text{ s}^{-1}, \quad (13)$$

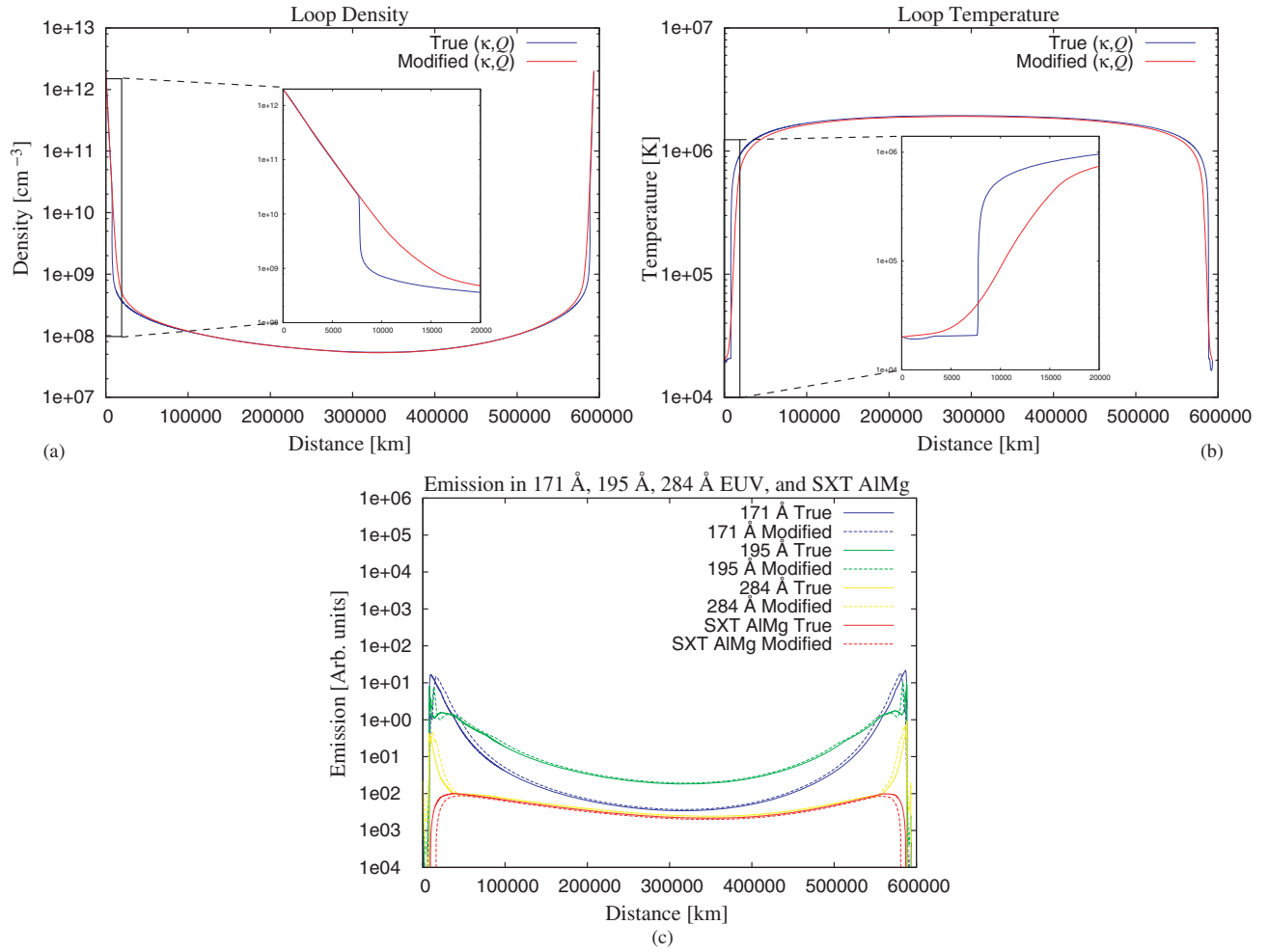


Figure 3. (a) Density and (b) temperature profiles in a one-dimensional loop with the third heating model of Section 2.2. The blue curves show a solution with *Spitzer* thermal conductivity, κ , and Athay's radiation loss function, Q , whereas the red curves show the solution when κ and Q are modified. Note that the two solutions agree very well in the corona, but that the modified (red) solution has a much broader transition region. (c) The emission along the loop for the two solutions, showing that the emissivity is accurately reproduced by the modified model at coronal temperatures (above 500,000 K). The location of the loop in the global corona is shown in Figure 2.

(A color version of this figure is available in the online journal.)

where B is the magnetic field strength at the base and L is the half-length of the loop. Despite its elegance and deceptive simplicity, the implementation of this model in a three-dimensional computation is not straightforward. First, we note that in principle the heat flux can become infinite as the length of the loops tends to zero near the neutral line. Hence, in calculating F_{ch} , a minimum loop-length criterion must be imposed. Second, as we will see in the discussion given below, the constant in front of the right-hand side applies only when using magnetic-flux distributions with the same resolution of the magnetogram of Schrijver et al. (2004) and generally must be rescaled at other resolutions. Third, this model was developed for one-dimensional calculations and requires field-line tracing to deposit the heat in the three-dimensional corona. Not only can this be a computationally intensive task, if it is done at every time step of a three-dimensional time-dependent calculation, but it also requires special care when calculating lengths of loops in regions where small initial errors may give large uncertainties in the final result (i.e., near quasi-separatrix layers, Titov et al. 2008).

To deposit heat in the volume for the Schrijver et al. (2004) heating specification, we used the following method: first,

a heat-flux map is constructed for each (θ, ϕ) grid point at the lower boundary from Equation (13). To deposit this heat in the calculation volume (i.e., to obtain H_{ch}), a field line is traced from each point on the heat-flux map and the grid cells intersected by this field line are identified. If the field line is closed (returns to the solar surface), then we deposit the following volumetric heating contribution into each intersected cell:

$$\Delta H_{\text{ch}}^{(j)} = \frac{F_{\text{ch}}^{(j)} \Delta A_j}{\sum_k \Delta V_k}, \quad (14)$$

where ΔA_j is the area associated with each point j on the heat-flux map and ΔV_k is the volume of the k th computational cells intersected by the j th field line. This corresponds to assuming uniform heating along each loop, as in Schrijver et al. (2004). Each cell volume may be intersected by zero, one, or several field lines and the total deposited heating at $\mathbf{x}_j$ is the sum of all contributions:

$$H_{\text{ch}}(\mathbf{x}_j) = \sum_j \Delta H_{\text{ch}}^{(j)}. \quad (15)$$

Since this formula applies only to closed-field regions and we are also modeling coronal holes, we have added to H_{ch}

the heating calculated with model 1. For computational simplicity, we did not recompute the heating at every time step, but used the magnetic field configuration obtained with heating model 1 and held the heating constant in time. While the emission was greatly affected by changes in the heating model, changes to the topology and structure of the magnetic field were relatively small, justifying this approximation.

In our first attempt to obtain the volumetric heating H_{ch} , we used the magnetic flux distribution of Figure 2(a) to calculate the heat flux F_{ch} according to Equation (13). With this method, the total power deposited in the corona was $3.77 \times 10^{27} \text{ erg s}^{-1}$, which turned out to be insufficient to heat the corona and gave very dim emission and a poor comparison with observations. The reason for this poor result is that the smoothed magnetic map used for the calculation has a smaller amount of unsigned magnetic flux than the maps used by Schrijver et al. (2004). If we chose a higher resolution magnetic map than Schrijver et al., we would in general obtain too high a heat flux. A further complication is that the heat flux depends on the accuracy of the field line tracing. A higher accuracy tracing routine (such as the one we use) captures more short-scale loops and yields a higher heat flux than obtained by Schrijver et al. (2004). To develop a heating that matched as closely as possible the Schrijver et al. heating, we obtained a heat-flux map from Dr. Schrijver that was developed for this time period, as shown in Figure 4(b). This was based on a source-surface model using a magnetic map for CR1913 with the same resolution used by Schrijver et al. (2004) and had an integrated total power of $2.65 \times 10^{28} \text{ erg s}^{-1}$. After computing H_{ch} with the method described above, the total power deposited in the volume was $2.41 \times 10^{28} \text{ erg s}^{-1}$. This does not exactly match the power obtained with Equation (13) because in general the open- and closed-field regions from the MHD solution are different from those in the source-surface model. In addition, to provide heating in the coronal hole regions, we have added to H_{ch} the heating calculated with model 1, bringing the total power to $2.90 \times 10^{28} \text{ erg s}^{-1}$. We have also added the same Alfvén wave pressure of model 1.

3. While the solutions that we show in this paper are found by integrating the equations to steady state, in general we are interested in investigating time-dependent solutions, for example for studying coronal mass ejections (CMEs). The magnetic field structure will then evolve considerably during the computation, and if we are using heating model 2, we must frequently recalculate the computationally expensive Equations (14) and (15). Therefore, we have developed an empirical heating model that depends only on local variables and can be computed rapidly. This third heating specification comprises three terms, the previous exponential heat function of Equation (12) plus two new terms representing the quiet Sun and active region heating:

$$H_{\text{ch}} = H_{\text{exp}} + H_{\text{QS}} + H_{\text{AR}}, \quad (16)$$

$$H_{\text{QS}} = H_{\text{QS}}^0 f(r) \frac{B_t^2}{B(|B_r| + B_r^c)}, \quad (17)$$

$$H_{\text{AR}} = H_{\text{AR}}^0 g(B) \left(\frac{B}{B_0} \right)^{1.2}, \quad (18)$$

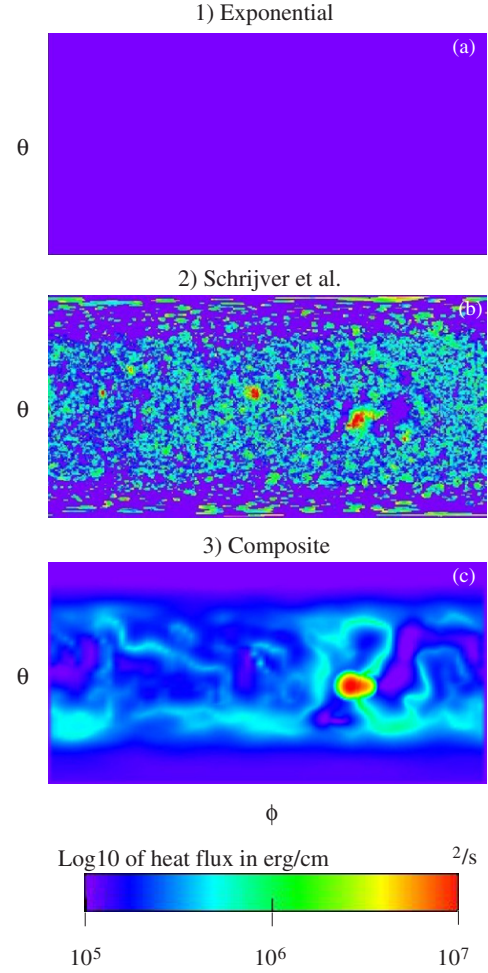


Figure 4. Heat flux for the three heating models described in Section 2.2: (a) exponential; (b) the model of Schrijver et al. (2004) calculated from a high-resolution magnetogram; and (c) composite model that includes terms to mimic the heating for the quiet Sun and for active regions.

(A color version of this figure is available in the online journal.)

where $B_t = \sqrt{B_\theta^2 + B_\phi^2}$ is the tangential magnetic field. For this case, the best results were obtained for $H_{\text{QS}}^0 = 1.18 \times 10^{-5} \text{ erg cm}^{-3} \text{ s}^{-1}$, $B_r^c = 0.55 \text{ G}$, $H_{\text{AR}}^0 = 1.87 \times 10^{-5} \text{ erg cm}^{-3} \text{ s}^{-1}$, and $B_0 = 1 \text{ G}$. The first term (H_{exp}) is necessary to accelerate the solar wind. However, we found that this term does not deposit enough energy to realistically heat the closed-field regions, in particular along neutral lines. To deposit more heat in the quiet Sun, a second term (H_{QS}) is introduced. This term mimics the $1/L$ behavior of heating in the quiet Sun, since it is large along neutral lines, where $B_r = 0$, and the short loops that surmount them. A third term (H_{AR}) depends on the strength of the magnetic field and adds more heating in active regions, where the magnetic field is larger and emission stronger. The dimensionless functions $f(r)$ and $g(B)$ are included in Equations (17) and (18) to limit the effect of the terms H_{QS} and H_{AR} to their intended regions. For this case,

$$f(r) = \frac{1}{2} \left(1 + \tanh \frac{1.7 - r/R_\odot}{0.1} \right) \exp \left(-\frac{r/R_\odot - 1}{0.2} \right),$$

$$g(B) = \frac{1}{2} \left(1 + \tanh \frac{B - 18.1}{3.97} \right),$$

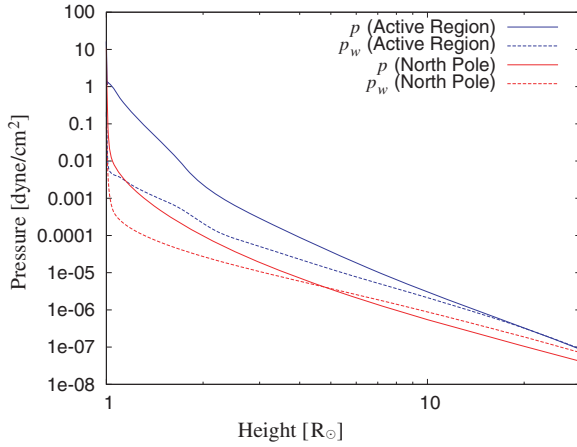


Figure 5. Comparison of the thermal (solid lines) and Alfvén wave pressure (dashed lines) as a function of height above the solar surface, starting from the active region (blue) and from the North Pole (red).

(A color version of this figure is available in the online journal.)

which serve to turn off the heating when $r \gtrsim 1.7R_\odot$ and $B \lesssim 18$ G, respectively. The same Alfvén wave pressure of model 1 has also been added to this model. The relative importance of Alfvén wave and thermal pressure can be appreciated in Figure 5, where they are plotted as a function of height above the surface, starting from the North Pole and from the main active region.

Although this model is still dependent on the magnetogram resolution as is heating model 2, it does not involve field-line tracing, making it suitable for three-dimensional computations.

2.3. Synthetic Emission Images

From the density and temperature calculated with our MHD model it is possible to produce synthetic emission images (Mok et al. 2005). The count registered by an instrument in a given configuration i is given by

$$D = \int n_e^2(w) f_i(T(w), n_e(w)) dw \quad [\text{DN s}^{-1} \text{ pixel}^{-1}], \quad (19)$$

where D is integrated along the line of sight w . The function $f_i(T, n_e)$ takes into account atomic physics, geometry, and the properties of both the instrument and the filters. Since the dependence of f_i on the electron density n_e is rather weak, we have neglected it in this investigation. The shapes of f_i for the EIT 171, 195, 284 Å filters and for the SXT AIMg configuration are shown in Figure 6. These are the configurations in which processed emission images may be obtained using the EIT processing software³ or from the *Yohkoh* Legacy Data Archive.⁴

3. RESULTS

Figure 2(a) shows the smoothed Kitt Peak synoptic map of B_r for CR1913 that was used at the lower boundary for all the calculations. A potential-field extrapolation has been used as the initial condition for the magnetic field, while the plasma temperature, density, and velocity were imposed from a one-dimensional solar-wind solution calculated previously.

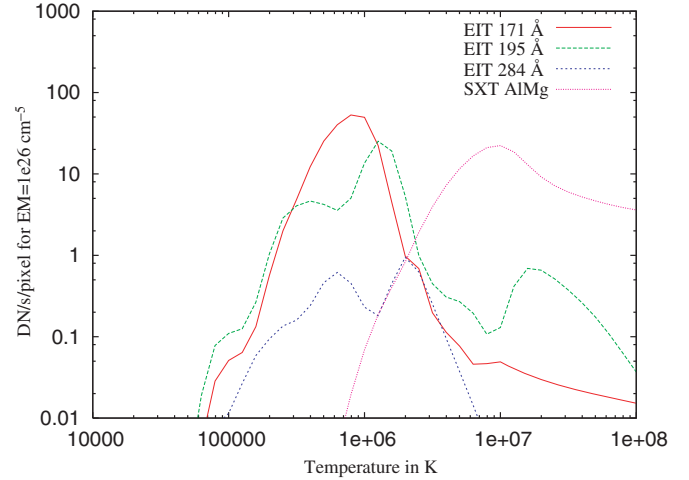


Figure 6. Response functions for EIT 171, 195, and 284 Å filters and for SXT AIMg filters used to calculate the synthetic emission images in this work. The curves are calculated for an emission measure $EM = \int n_e^2 dw = 10^{26} \text{ cm}^{-5}$.

(A color version of this figure is available in the online journal.)

We have initially prescribed heating model 1 and have advanced Equations (3)–(7) until a steady state with coronal holes, streamers, and the heliospheric current sheet is reached. Figure 2(b) shows the magnetic field lines in the corona: open-field lines are visible especially in the polar regions, while closed-field lines are present at equatorial latitudes, crowned by the characteristic cusp-like field lines. We then restarted the computation, but using heating model number 2 instead, and let the corona relax to another steady state. This has an important effect on the plasma properties, but leaves the magnetic field almost unchanged. We repeated this procedure (changing of the heating model, followed by relaxation) for heating model 3. The relaxation phase was in all cases at least 1 day. We found that this is considerably longer than the time required for the emission in the lower corona to settle down to unchanging values.

The dots in Figure 2(a) mark the footpoints of the field line that has been used to calculate the one-dimensional loop solutions shown in Figures 1 and 3. The three-dimensional aspect of this loop can be seen in Figure 2(b), where it is indicated with an arrow. In Figure 7, we show a comparison between the density and temperature extracted along the same loop from the three-dimensional solution and the corresponding one-dimensional calculation. As can be seen in the figure, the solutions are nearly identical. In the one-dimensional solution, thermal conduction is exactly parallel to the magnetic field. Figure 7 demonstrates that whatever perpendicular thermal conduction is introduced in the three-dimensional solution by numerical errors, it does not significantly affect the solution.

Using the density and temperature obtained with the three heating models described in Section 2.2 and calculated at the end of the relaxation phase, we prepared the synthetic emission images shown in Figure 8. We computed the emission in the following bands/configurations: EIT 171, 195, and 284 Å, and SXT AIMg. Figure 8 also displays the corresponding observations to allow a quantitative comparison with the models. In Figure 9, the calculated and observed emissions in the aforementioned bands are evaluated and compared with the observed $\text{DN s}^{-1} \text{ pixel}^{-1}$ along line cuts at the solar equator. Another cut, along the north-south direction and intersecting the active region, is presented in Figure 10.

³ http://umbra.nascom.nasa.gov/eit/eit_guide/

⁴ <http://solar.physics.montana.edu/ylegacy/>

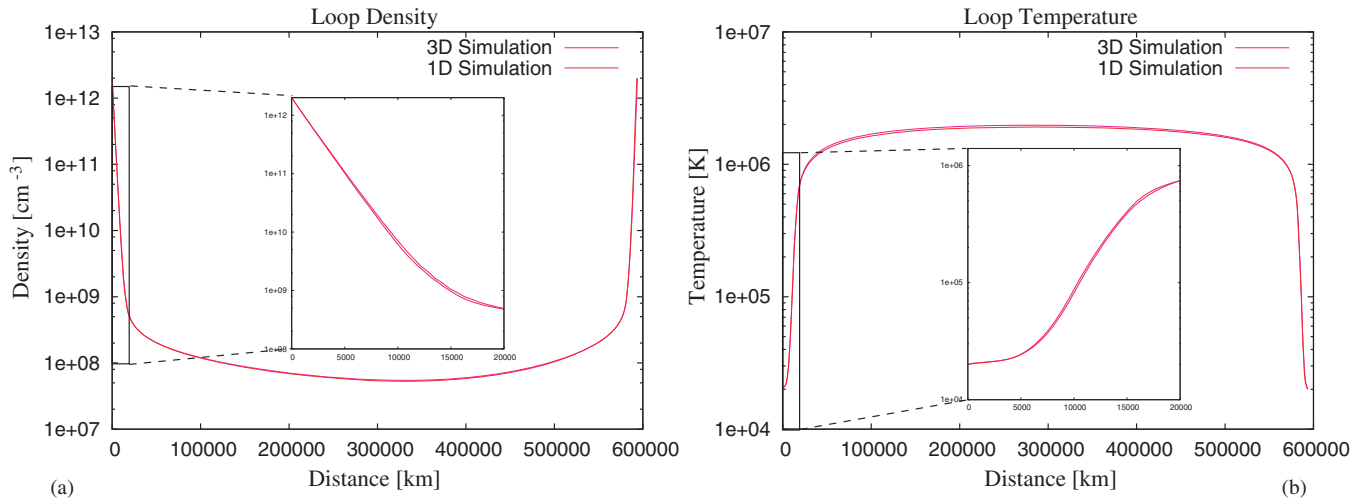


Figure 7. (a) Density and (b) temperature profiles in a one-dimensional loop with the third heating model of Section 2.2. The magenta curves show quantities extracted from the three-dimensional MHD calculation, whereas the red curves show the solution calculated with a one-dimensional HD model. The location of the loop in the global corona is shown in Figure 2.

(A color version of this figure is available in the online journal.)

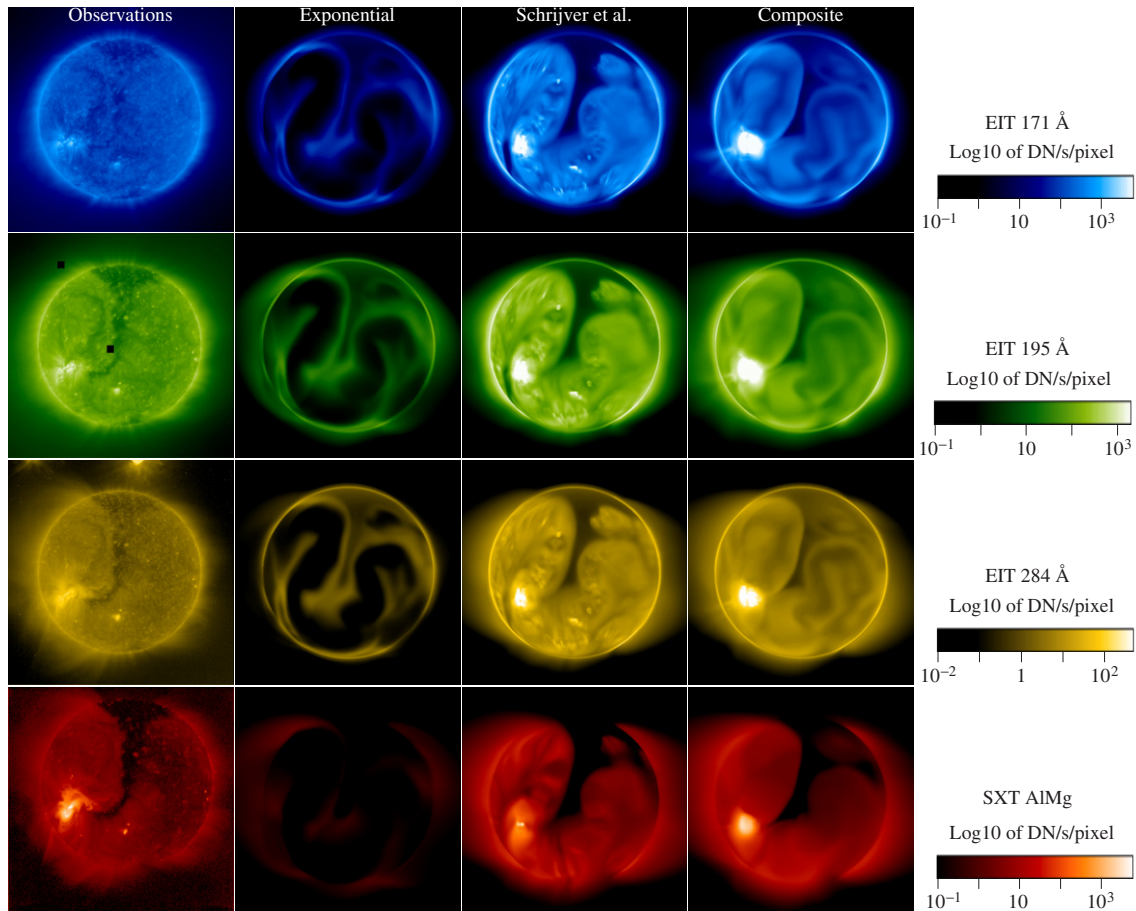


Figure 8. Comparison of observed and synthetic emission images. The first column shows the observed emission, each remaining column shows the computed emission for models 1, 2, and 3 (see Section 2.2). Rows show emission in the EIT 171, 195, and 284 Å band and in the SXT AlMg configuration. The observations were taken on 1996 August 27 at 00:00:13, 00:24:05, 01:05:19, and 02:11:33 UT, respectively. The resolution is 1024^2 pixels for the EIT images and 512^2 pixels for the SXT. The computed emission images were calculated for 00:30:00 UT, corresponding to a central meridian Carrington longitude of 296° .

(A color version of this figure is available in the online journal.)

Figures 8–10 show that heating model 1 (“exponential” heating), used here as a reference model, is clearly inadequate for describing emission in the corona. Therefore, heating suitable

for generating fast wind from coronal holes is not sufficient for powering emission in the quiet Sun and active regions. Note the large dark areas in the closed-field region that appear for this

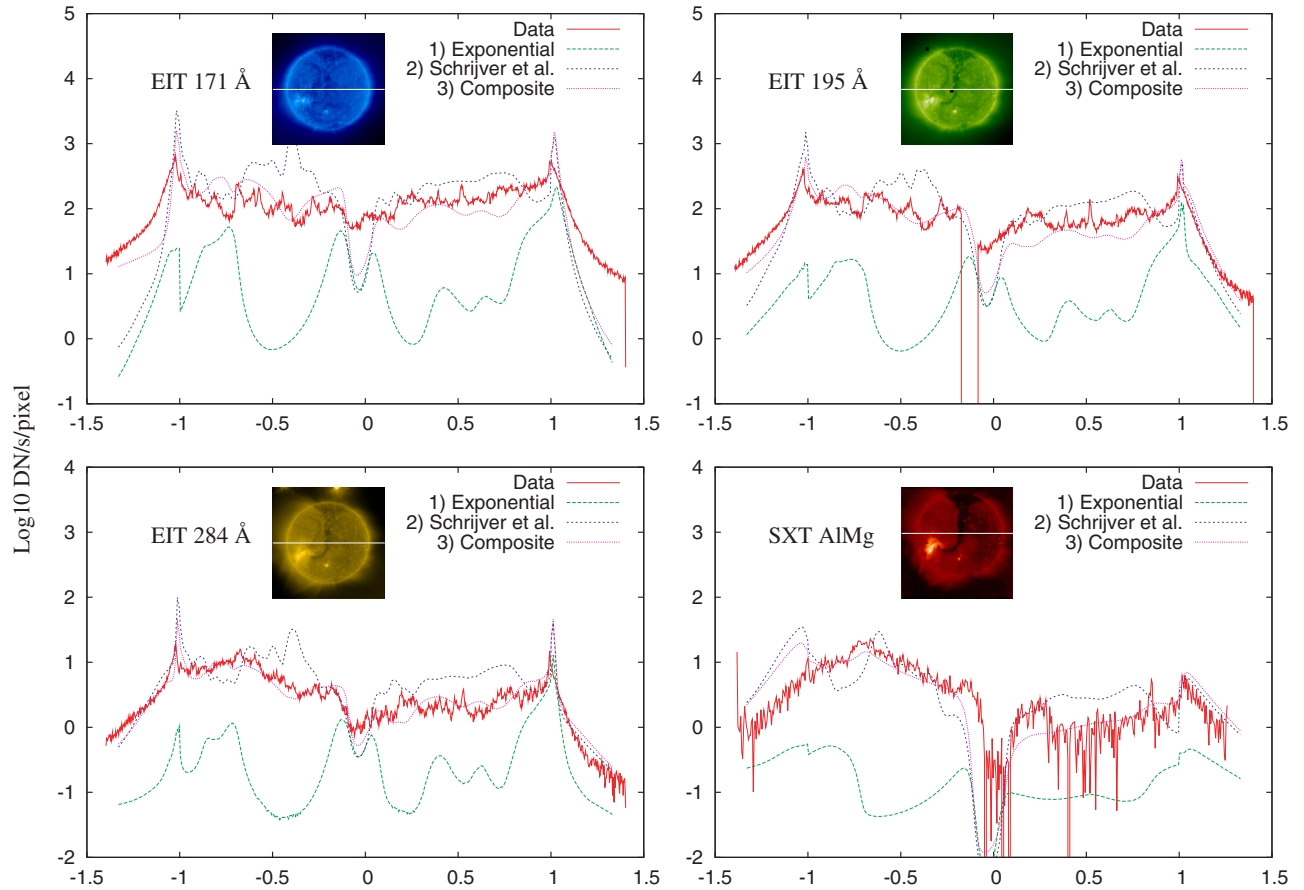


Figure 9. Comparison of observed and synthetic emission along an equatorial cut. The synthetic emission was obtained using models 1, 2, and 3 of Section 2.2. (A color version of this figure is available in the online journal.)

heating model; these follow the large-scale magnetic-polarity inversion line, and demonstrate that shorter loops require more heat-per-unit-volume than longer loops in order to reproduce observed emission. This result is, in fact, embedded in the scaling law of Rosner et al. (1978), and explains the requirement of the $1/L$ term in heating model 2 and the H_{QS} term in heating model 3, which is larger in the shorter loops surmounting neutral lines.

Figures 8–10 show that heating models 2 and 3 capture many of the features of the observed emission. The overall structure of the corona, including the “elephant’s trunk” corona hole, is clearly present. Quantitatively, the levels of EUV and X-ray emission in the quiet Sun, and X-ray emission in the active region, are also reproduced approximately in the simulations. The results for heating model 3 are closest to the observed emission in Figures 9 and 10, while model 2 is slightly too high. Heating model 2 yields more fine-scale structures that match some of the observed structures in Figure 8.

We also note discrepancies between the observed emission and simulated emission for models 2 and 3. The extended coronal hole covers a larger area in the models than in the observations. This is probably due to the smoothing of the magnetogram used as boundary condition for the magnetic field. This is also the likely reason for the size of the active region in the models being larger than in the observations. The vertical cuts in Figure 10 show that emission in the active region for models 2 and 3 matches the SXT observation, but it is too high in EUV. Emission on the limbs is reproduced fairly well by models 2 and 3 in the east–west cuts of Figure 9 for EIT 195 Å, 284 Å,

and SXT. It appears too low in EIT 171 Å: this seems to be in agreement with the results of Aschwanden et al. (2001), who found that steady-state models do not reproduce correctly loop emission but they only show bright end points, thus indicating that dynamic solutions may be necessary to improve the match (Mok et al. 2008). In the north–south cuts of Figure 10 the comparison with observations of the limb EUV emission is even less successful: observed emission in the northern and southern limbs is higher than in the models, as it is also evident from Figure 8. We also note that the simulated EUV emission from the models in the center of the coronal hole in the east–west cut (Figure 9) is also much less than that observed. We believe that these discrepancies are related; a shorter length scale heating may be present in these regions that is not included in the heating models.

4. DISCUSSION

We have shown that by using a three-dimensional MHD-thermodynamic algorithm it is possible to reproduce many of the multispectral emissive properties of the corona in a simulation that self-consistently includes the solar wind, open and closed structures, and the heliospheric current sheet. The values of the coronal emission of CR1913 have been compared quantitatively with observations, thus providing a strong empirical constraint on coronal heating models. We emphasize that our method does not allow us to infer the quantitative nature of coronal heating, but it allows us to quantitatively compare the predicted emission from an assumed coronal heating model with the emission

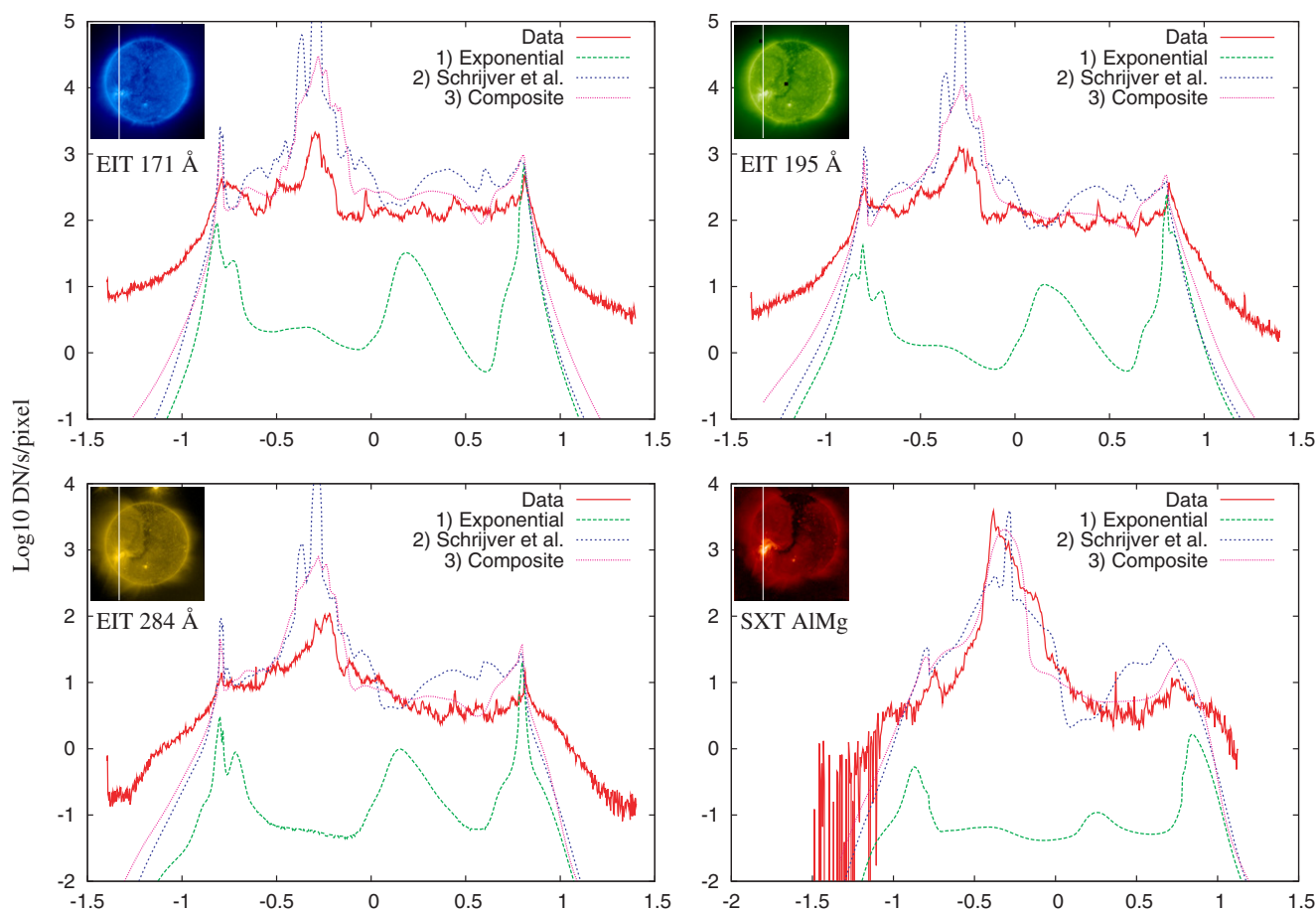


Figure 10. Comparison of observed and synthetic emission along a cut intersecting the active region. The synthetic emission was obtained using models 1, 2, and 3 of Section 2.2.

(A color version of this figure is available in the online journal.)

obtained from observations. To illustrate our technique, we have analyzed the emission from three heating models. Figures 8–10 show that none of the heating models give a perfect match with observations. However, among the models considered here, model 1 clearly appears to be inadequate: although it provides adequate acceleration to the solar wind, the heating is insufficient to reproduce the quiet Sun emission. The heating model (2) of Schrijver et al. (2004) and the composite heating model (3) give a reasonable match to observations.

While the model of Schrijver et al. (2004) yields a realistic rendition of the corona, it is difficult to correctly apply in three-dimensional MHD simulations, especially if the magnetic structure evolves in time. We have developed heating model 3 as an alternative that can be computed efficiently in a time-dependent MHD computation. This heating model performs slightly better than model 2 in quantitatively predicting the quiet Sun emission in EUV and X-rays. Both models do a reasonable job of estimating the peak X-ray emission from the active region. Both heating models have similar disagreements with the observations, as described in Section 3. In particular, the EUV emission in the active region is too large. Long loops emit less in EUV than what is observed, the brightness being confined at the footpoints (this is particularly evident at the limbs). This mismatch has been discussed by Klimchuk (2006) and was also found by Mok et al. (2008), who concluded that dynamic solutions may be necessary to reproduce the observations. Therefore, while our results show that three-dimensional MHD simulations have

advanced to the point that they can begin to quantitatively predict the properties of the corona over a range of temperatures, there is still considerable room for improvement in the heating models. Our calculations have shown how it is possible to evaluate the accuracy of empirical heating models by comparing observations with three-dimensional MHD solutions. In principle, this technique could also be used to evaluate theoretical models of coronal heating, providing that the models are sufficiently advanced to quantitatively predict heating in detail on macroscopic scales. The relevance of this study is not limited to the long-standing problem of coronal heating. In fact, realistic background solutions are necessary to model the dynamic activity of the corona (e.g., filament formation, flares, and CMEs) in the context of real solar events. These studies are currently under way.

We are grateful to Drs. Carolus Schrijver and Marc DeRosa for in-depth discussions about their model and for supplying the heat flux map shown in Figure 4. We also thank Drs. Yung Mok and Amy Winebarger for useful discussions. This work was supported by AFOSR, by the NASA LWS, HTP, and strategic capabilities programs, and by NSF through the Center for Integrated Space Weather Modeling.

REFERENCES

- Altschuler, M. D., & Newkirk, G. 1969, *Sol. Phys.*, **9**, 131
 Aschwanden, M. J. 2001, *ApJ*, **560**, 1035

- Aschwanden, M. J., Schrijver, C. J., & Alexander, D. 2001, *ApJ*, **550**, 1036
- Athay, R. G. 1986, *ApJ*, **308**, 975
- Bray, R. J., Cram, L. E., Durrant, C., & Loughhead, R. E. 1991, in *Plasma Loops in the Solar Corona* ed. R. J. Bray, L. E. Cram, C. Durrant, & R. E. Loughhead (Cambridge: Cambridge Univ. Press), 522
- Cheng, C.-C., Oran, E. S., Doschek, G. A., Boris, J. P., & Mariska, J. T. 1983, *ApJ*, **265**, 1090
- Cohen, O., et al. 2007, *ApJ*, **654**, L163
- Craig, I. J. D., McClymont, A. N., & Underwood, J. H. 1978, *A&A*, **70**, 1
- Dere, K. P., Landi, E., Mason, H. E., Monsignori Fossi, B. C., & Young, P. R. 1997, *A&AS*, **125**, 149
- Endeve, E., Holzer, T. E., & Leer, E. 2004, *ApJ*, **603**, 307
- Endeve, E., Leer, E., & Holzer, T. E. 2003, *ApJ*, **589**, 1040
- Endler, F. 1971, *Mitt. Astron. Ges. Hamburg*, **30**, 136
- Fisher, G. H., Longcope, D. W., Metcalf, T. R., & Pevtsov, A. A. 1998, *ApJ*, **508**, 885
- Gudiksen, B. V., & Nordlund, Å. 2005a, *ApJ*, **618**, 1031
- Gudiksen, B. V., & Nordlund, Å. 2005b, *ApJ*, **618**, 1020
- Gurman, J. B., Withbroe, G. L., & Harvey, J. W. 1974, *Sol. Phys.*, **34**, 105
- Hayashi, K. 2005, *ApJS*, **161**, 480
- Hoeksema, J. T., Wilcox, J. M., & Scherrer, P. H. 1983, *J. Geophys. Res.*, **88**, 9910
- Hollweg, J. V. 1978, *Rev. Geophys. Space Phys.*, **16**, 689
- Hood, A. W., & Priest, E. R. 1979, *A&A*, **77**, 233
- Jacques, S. A. 1977, *ApJ*, **215**, 942
- Karpen, J. T., Antiochos, S. K., Hohensee, M., Klimchuk, J. A., & MacNeice, P. J. 2001, *ApJ*, **553**, L85
- Keppens, R., & Goedbloed, J. P. 1999, *A&A*, **343**, 251
- Klimchuk, J. A. 2006, *Sol. Phys.*, **234**, 41
- Klimchuk, J. A., Antiochos, S. K., & Mariska, J. T. 1987, *ApJ*, **320**, 409
- Li, B., Li, X., Hu, Y.-Q., & Habbal, S. R. 2004, *J. Geophys. Res. (Space Phys.)*, **109**, 7103
- Linker, J. A., & Mikić, Z. 1995, *ApJ*, **438**, L45
- Linker, J. A., van Hoven, G., & Schnack, D. D. 1990, *Geophys. Res. Lett.*, **17**, 2281
- Linker, J. A., et al. 1999, *J. Geophys. Res.*, **104**, 9809
- Lionello, R., Linker, J. A., & Mikić, Z. 2001, *ApJ*, **546**, 542
- Lundquist, L. L., Fisher, G. H., McTiernan, J. M., & Régnier, S. 2004, in *ESA SP-575: SOHO 15 Coronal Heating*, ed. R. W. Walsh, J. Ireland, D. Danesy, & B. Fleck (Noordwijk: ESA), 306
- Mandrini, C. H., Démoulin, P., & Klimchuk, J. A. 2000, *ApJ*, **530**, 999
- Mikić, Z., & Linker, J. A. 1996, in *AIP Conf. Proc.*, Vol. 382, *Solar Wind Eight*, Proc. Eighth Int. Solar Wind Conf., ed. D. Winterhalter, J. T. Gosling, S. R. Habbal, W. S. Kurth, & M. Neugebauer (Woodbury, NY: AIP), 104
- Mikić, Z., Linker, J. A., Schnack, D. D., Lionello, R., & Tarditi, A. 1999, *Phys. Plasmas*, **6**, 2217
- Mok, Y., Mikić, Z., Lionello, R., & Linker, J. A. 2005, *ApJ*, **621**, 1098
- Mok, Y., Mikić, Z., Lionello, R., & Linker, J. A. 2008, *ApJ*, **679**, L161
- Mok, Y., Schnack, D. D., & van Hoven, G. 1991, *Sol. Phys.*, **132**, 95
- Neugebauer, M., et al. 1998, *J. Geophys. Res.*, **103**, 14587
- Parker, E. N. 1979, *Cosmical Magnetic Fields: Their Origin and Their Activity* (Oxford/New York: Clarendon/Oxford Univ. Press), 858
- Pevtsov, A. A., Fisher, G. H., Acton, L. W., Longcope, D. W., Johns-Krull, C. M., Kankelborg, C. C., & Metcalf, T. R. 2003, *ApJ*, **598**, 1387
- Pneuman, G. W., & Kopp, R. A. 1971, *Sol. Phys.*, **18**, 258
- Priest, E. R., Foley, C. R., Heyvaerts, J., Arber, T. D., Mackay, D., Culhane, J. L., & Acton, L. W. 2000, *ApJ*, **539**, 1002
- Riley, P., Linker, J. A., Mikić, Z., Lionello, R., Ledvina, S. A., & Luhmann, J. G. 2006, *ApJ*, **653**, 1510
- Rosner, R., Tucker, W. H., & Vaiana, G. S. 1978, *ApJ*, **220**, 643
- Roussev, I. I., et al. 2003, *ApJ*, **595**, L57
- Schatten, K. H. 1971, *Cosm. Electrodyn.*, **2**, 232
- Schatten, K. H., Wilcox, J. M., & Ness, N. F. 1969, *Sol. Phys.*, **6**, 442
- Schrijver, C. J., Sandman, A. W., Aschwanden, M. J., & DeRosa, M. L. 2004, *ApJ*, **615**, 512
- Schrijver, C. J., Zwaan, C., Maxson, C. W., & Noyes, R. W. 1985, *A&A*, **149**, 123
- Serio, S., Peres, G., Vaiana, G. S., Golub, L., & Rosner, R. 1981, *ApJ*, **243**, 288
- Sittler, E. C., Ofman, L., Gibson, S., Guhathakurta, M., Davila, J., Skoug, R., Fludra, A., & Holzer, T. 2003, in *AIP Conf. Ser.*, Vol. 679, *Solar Wind Ten*, ed. M. Velli, R. Bruno, F. Malara, & B. Bucci (New York: AIP), 113
- Steinolfson, R. S., Suess, S. T., & Wu, S. T. 1982, *ApJ*, **255**, 730
- Suess, S. T., Wang, A.-H., & Wu, S. T. 1996, *J. Geophys. Res.*, **101**, 19957
- Suess, S. T., Wang, A.-H., Wu, S. T., Poletto, G., & McComas, D. J. 1999, *J. Geophys. Res.*, **104**, 4697
- Titov, V. S., Mikić, Z., Linker, J. A., & Lionello, R. 2008, *ApJ*, **675**, 1614
- Usmanov, A. V. 1993, *Sol. Phys.*, **146**, 377
- Usmanov, A. V. 1996, in *AIP Conf. Proc.* 382, *Solar Wind Eight*, ed. D. Winterhalter, J. T. Gosling, S. R. Habbal, W. S. Kurth, & M. Neugebauer (Woodbury, NY: AIP), 65
- Usmanov, A. V., & Goldstein, M. L. 2003, *J. Geophys. Res. (Space Phys.)*, **108**, 1
- Vesecky, J. F., Antiochos, S. K., & Underwood, J. H. 1979, *ApJ*, **233**, 987
- Wang, Y.-M., & Sheeley, N. R., Jr. 1988, *J. Geophys. Res.*, **93**, 11227
- Wang, Y.-M., & Sheeley, N. R., Jr. 1995, *ApJ*, **447**, L143
- Wang, A.-H., Wu, S. T., Suess, S. T., & Poletto, G. 1993, *Sol. Phys.*, **147**, 55
- Warren, H. P., & Winebarger, A. R. 2006, *ApJ*, **645**, 711
- Washimi, H., Ogino, T., & Yoshino, Y. 1987, *Geophys. Res. Lett.*, **14**, 487
- Withbroe, G. L. 1988, *ApJ*, **325**, 442
- Wu, S. T., Kan, L. C., Nakagawa, Y., & Tandberg-Hanssen, E. 1981, *Sol. Phys.*, **70**, 137
- Zhao, X., & Hoeksema, J. T. 1995, *J. Geophys. Res.*, **100**, 19